Task-Dependent Optimal Linear Observers in Linear Gaussian Dynamical Systems

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For a stationary linear Gaussian dynamical system, the rank-r linear observer minimizing squared-error prediction loss for target $Cx_{t+\tau}$ has row space equal to the top-r eigenspace of the target information matrix $M_{C,\tau} = \Sigma^{1/2}(A^\tau)^\top C^\top C A^\tau \Sigma^{1/2}$, uniquely up to eigenspace rotation. Two targets with distinct dominant eigenspaces admit no common optimal rank-r observer. A common optimal observer exists if and only if the target family is r-*coherent*: all information matrices share a common top-r eigenspace. This condition is exact, not asymptotic or approximate. Numerical illustrations at $n = 8$, $r = 2$ confirm a target-dependent case ($\Gamma = 0.026 > 0$) and an r-coherent construction, both consistent with the theorem.

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I. INTRODUCTION

A central question in observer design for linear dynamical systems is whether a single rank-r linear observer can be simultaneously optimal for predicting multiple output channels $\{C_k x_{t+\tau_k}\}$, or whether the optimal observer depends on the specific prediction target and horizon. For finite-dimensional linear Gaussian dynamical systems (LGDS) at stationarity, this paper gives an exact answer: the optimal rank-r observer for predicting $Cx_{t+\tau}$ is the top-r eigenspace of the target information matrix $M_{C,\tau}$, and a common optimal observer across a family of targets exists if and only if all information matrices share a common dominant eigenspace.

This paper is not about dynamic observer synthesis in the Luenberger–Kalman sense [1–3]: the observer is a fixed rank-r linear map $\phi_U(x) = U\Sigma^{-1/2}x$, and performance is squared-error prediction loss for a specified target $Cx_{t+\tau}$ at stationarity. The static case ($A = I$, $\tau = 0$) recovers standard principal-subspace selection, but the main object here is the interaction of system dynamics, prediction horizon τ, and observer rank through the information matrix $M_{C,\tau}$.

Reduced representations for linear systems have been studied via principal components of Gramians [4], balanced truncation, and related model-reduction methods [5]. Those works address reduced system descriptions or approximation error across all inputs and initial conditions. The present note asks a different question: given a prediction task family with possibly distinct information matrices $\{M_{C_k,\tau_k}\}$, when does one rank-r observer remain exactly optimal for all targets, and when is target-dependent observer selection unavoidable?

The Grassmannian $\mathrm{Gr}(r, n)$ is the natural parameter space for rank-r linear observers [6], and observer quality—the trace $\mathrm{tr}(UM_{C,\tau}U^\top)$—is a smooth function on this manifold. Optimal observer selection reduces to trace maximisation on $\mathrm{Gr}(r, n)$, governed by the Ky Fan variational principle [7]: the maximum is achieved by the leading eigenspace of $M_{C,\tau}$, recovering principal-subspace selection [8] in the static ($A = I$) case within the classical linear state-space framework [3].

The Ky Fan variational principle and principal-subspace optimality are classical [7, 8]. The contribution here is the exact algebraic biconditional for the existence of a common optimal rank-r observer across a prediction task family: r-coherence (Definition 3), stated precisely in Corollary 4. The regret gap Γ provides a computable certificate of unavoidable suboptimality when r-coherence fails.

The result is scoped to finite-dimensional LGDS, fixed-rank linear observers, and squared-error prediction targets.

II. SYSTEM AND OBSERVER FAMILY

a. Dynamics. Let

$$x_{t+1} = Ax_t + \xi_t, \quad \xi_t \sim \mathcal{N}(0, Q), \quad x_t \in \mathbb{R}^n, \tag{1}$$

with A stable (spectral radius < 1) and $Q \succ 0$. The unique stationary covariance Σ satisfies $A\Sigma A^\top + Q = \Sigma$ and is strictly positive definite [3].

b. Observer family. An observer of rank r ($1 \le r < n$) is parametrised by a row-orthonormal matrix $U \in \mathbb{R}^{r\times n}$, $UU^\top = I_r$:

$$\phi_U(x) = U\Sigma^{-1/2}x. \tag{2}$$

Pre-whitening by $\Sigma^{-1/2}$ places all observers on a common isotropic footing; in the static case ($A = I$, $\tau = 0$) this reduces to principal component analysis [8]. Two observers differing only by an orthogonal rotation of their row spaces are equivalent; the observer space is the Grassmannian $\mathrm{Gr}(r, n)$, a smooth compact Riemannian manifold of dimension $r(n-r)$, with metric and geodesic structure from $O(n)$ [6].

c. Target family. A prediction target is a pair (C, τ) with $C \in \mathbb{R}^{p\times n}$ and integer $\tau \ge 1$:

$$y_{C,\tau} = Cx_{t+\tau}. \tag{3}$$

Performance is total mean squared prediction error. A target family $\mathcal{T} = \{(C_k, \tau_k)\}$ is any finite collection.

III. THEOREM

A. Exact Bayes risk formula

Given observation $z = U\Sigma^{-1/2}x_t$, the MMSE predictor of $Cx_{t+\tau}$ is linear by Gaussian conjugacy. At stationarity, $\mathrm{Cov}(Cx_{t+\tau}, z^\top) = CA^\tau\Sigma^{1/2}U^\top$ and $\mathrm{Cov}(z, z^\top) = I_r$. The Bayes risk is

$$\mathcal{R}(U;\, C, \tau) = \mathrm{tr}(C\Sigma C^\top) - \mathrm{tr}\big(U\, M_{C,\tau}\, U^\top\big), \tag{4}$$

where the *target information matrix* is

$$M_{C,\tau} = \Sigma^{1/2}(A^\tau)^\top C^\top C A^\tau \Sigma^{1/2} \;\in\; \mathbb{R}^{n\times n}. \tag{5}$$

$M_{C,\tau}$ is symmetric positive semidefinite of rank at most $\min(p, n)$. The formula is exact.

B. Top-eigenspace optimality

Proposition 1. *For fixed rank r, the observer minimising $\mathcal{R}(U;\, C, \tau)$ over $\mathrm{Gr}(r, n)$ has row space equal to the top-r eigenspace of $M_{C,\tau}$. The minimum risk is*

$$\mathcal{R}^* = \mathrm{tr}(C\Sigma C^\top) - \sum_{i=1}^{r} \lambda_i(M_{C,\tau}), \tag{6}$$

where $\lambda_1 \geq \cdots \geq \lambda_n$ are the eigenvalues of $M_{C,\tau}$. Uniqueness up to rotation holds when $\lambda_r > \lambda_{r+1}$.

Degeneracy at the gap ($\lambda_r = \lambda_{r+1}$) yields a continuum of optimal observers spanning the degenerate eigenspace; in practice, a generic target (C, τ) produces a strict spectral gap, making degeneracy non-generic for fixed (A, Σ).

Proof. Minimising $\mathcal{R}$ is equivalent to maximising $\mathrm{tr}(UM_{C,\tau}U^\top)$ subject to $UU^\top = I_r$. By the Ky Fan variational principle [7, 9] the maximum equals $\sum_{i=1}^{r} \lambda_i(M_{C,\tau})$, achieved exactly when the row space of U spans the corresponding eigenspace. ■

C. Target-dependent observer selection

Corollary 2. *If targets (C_1, τ_1) and (C_2, τ_2) have information matrices M_1 and M_2 with distinct top-r eigenspaces, no single rank-r observer is simultaneously optimal for both.*

Proof. Optimality for target k requires the row space of U to equal the top-r eigenspace of M_k. Distinct eigenspaces cannot be spanned by the same U. ■

The corollary extends to any finite target family: if $\{M_k\}$ do not share a common top-r eigenspace, no common optimal rank-r observer exists.

D. r-coherence: the exact exception

Definition 3. *A target family $\mathcal{T}$ is* r-coherent *with respect to (A, Σ) if all matrices $\{M_{C,\tau} : (C, \tau) \in \mathcal{T}\}$ share a common top-r eigenspace V^*.*

Corollary 4. *A common optimal observer exists if and only if $\mathcal{T}$ is r-coherent.*

r-coherence requires all targets to be functions of a common r-dimensional sufficient statistic across all prediction horizons—a strong alignment between the system dynamics and the task family. Outside r-coherence, the optimal observer is target-dependent.

Remark 5. *r-coherence is an algebraic equality between eigenspaces, not an approximate or asymptotic condition. It is the exact condition for a common optimal observer to exist.*

IV. NUMERICAL ILLUSTRATIONS

The theorem is analytically exact under its stated assumptions. We illustrate it numerically with two finite-dimensional instances: a target-dependent case and an r-coherent case. All reported quantities are computed exactly from the analytic Bayes risk formula and numerical optimization on the Grassmannian.

a. Setup. State dimension $n = 8$, observer rank $r = 2$. The model is a finite-dimensional linear Gaussian dynamical system [10]. $A = VDV^{-1}$ with eigenvalues drawn from $U[0.3, 0.85]$, random non-orthogonal V (seed 100), $Q = 0.1\, I_8$. Σ is computed from the discrete Lyapunov equation; $\Sigma \succ 0$ is confirmed to machine precision. All reported quantities are computed exactly from the analytic Bayes risk formula, eigenspace calculations, and repeated projected optimization over $\mathrm{Gr}(2, 8)$ (300 random initializations of projected L-BFGS-B, gradient-norm threshold 10^{-12}, function-value tolerance 10^{-15}).

b. Condition A: target-dependent instance. Targets $T_1 = ([e_1^\top;\, e_2^\top],\, 1)$ and $T_2 = ([e_5^\top;\, e_6^\top],\, 5)$.

With non-normal A, A^1 and A^5 rotate state-space directions differently; the principal angles [11] between the top-2 eigenspaces of M_1 and M_2 are $54.1°$ and $24.0°$, confirming that no single rank-2 observer is simultaneously optimal.

The minimax regret gap

$$\Gamma = \min_{U \in \mathrm{Gr}(r,n)} \max_k \left[\mathcal{R}(U;\, C_k, \tau_k) - \mathcal{R}_k^*\right] \tag{7}$$

quantifies the best achievable worst-case suboptimality under task conflict. The minimax observer U_{joint} achieves $\Gamma = 0.026 > 0$. No common optimal rank-r observer exists for this target pair. The positive gap certifies the impossibility of any jointly optimal rank-2 observer for this task pair. Table I gives the full regret structure. The balanced regrets at U_{joint} (0.026 on both targets) are a necessary condition for a minimax with $\Gamma > 0$, confirming

TABLE I. Regret matrix, Condition A ($n = 8$, $r = 2$, seed 100). All values from the exact analytic Bayes risk formula.

Observer	Regret on T_1	Regret on T_2
U_1^* (optimal for T_1)	0.000	0.036
U_2^* (optimal for T_2)	0.763	0.000
U_{joint} (compromise)	0.026	0.026

the global minimum was found. The large asymmetry in the off-diagonal entries (0.763 vs. 0.036) reflects the principal angle structure: the larger 54.1° principal angle shows U_2^*'s subspace is substantially misaligned with M_1's top eigenspace, forfeiting most of its eigenvalue weight, while the smaller 24.0° angle means U_1^* retains partial alignment with M_2's top eigenspace.

c. Condition B: structurally coherent example. A target family is constructed so that all M_k share the same top-2 eigenspace by design: normal $A_{\text{sym}} = VDV^\top$ with orthogonal V, and a common C whose rows are the top-2 eigenvectors of A, varying only the prediction horizon $\tau \in \{1, 2, 3, 5\}$. Under this construction $M_{C,\tau}$ has the same dominant 2-dimensional eigenspace for all τ, so the four optimal observers are identical by the theorem. The maximum pairwise principal angle across all four is 0.000° to machine precision.

Under $\varepsilon = 0.1$ perturbations to C (20 random draws), the maximum principal angle across all draws and all pairs is 0.102 rad—well below the fragility threshold of 0.15 rad, set at 1.5ε. The threshold 1.5ε is a heuristic calibration; the direct numerical check confirms that the shared dominant eigenspace is robust to this level of perturbation in C. Structural coherence is perturbation-robust, distinguishing it from accidental near-coincidences of eigenspaces that break under small perturbations.

The numerical results are consistent with the theorem in all cases.

V. SCOPE

a. What is earned. In stationary finite-dimensional LGDS, among rank-r linear observers $\phi_U(x) = U\Sigma^{-1/2}x$ and squared-error prediction targets $y = Cx_{t+\tau}$, no common optimal rank-r observer exists unless the target family is r-coherent. r-coherence requires all prediction targets to be functions of a common r-dimensional sufficient statistic across all horizons—a strong alignment between the system dynamics and the task family. Outside r-coherence, no common optimal observer exists, and the cost of applying any fixed observer to a non-native target is given exactly by the Bayes risk formula. The Grassmannian $\mathrm{Gr}(r, n)$ is the literal domain of this optimisation: a compact Riemannian manifold on which observer quality is a smooth function, with $O(n)$ acting by transformation. The computation reduces to a single $n \times n$ eigendecomposition of $M_{C,\tau}$, independent of τ once A^τ is formed; A^τ is obtained in $O(n^3 \log \tau)$ operations by repeated squaring.

b. What is not earned. This result concerns optimal reduced linear observation for prediction; it does not address dynamic observer synthesis, asymptotic state reconstruction, or feedback design. It also does not apply to nonlinear systems, non-Gaussian noise, nonlinear observers, or non-prediction tasks. The breakdown is structural: for non-Gaussian noise or nonlinear dynamics, the MMSE predictor is generally nonlinear and Proposition 1's trace maximisation has no closed-form counterpart; for nonlinear observers, the Grassmannian parametrisation and the Ky Fan principle both cease to apply. The causal axis—whether the observer optimal for intervention differs from the observer optimal for prediction—is not addressed here.

c. Extensions. The linear Gaussian setting admits closed-form analysis because the MMSE predictor is linear, the stationary distribution is Gaussian, and the optimal observer reduces to a single eigendecomposition. Whether analogous shared-eigenspace conditions characterise common optimal observers in non-Gaussian or nonlinear systems, and whether intervention objectives impose additional constraints on observer selection, are open questions beyond the scope of the present result.

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